

Mukesh Reddy Mavurapu

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SUMMARY

Data Analyst with 5 years of experience in data analysis, BI reporting, ETL development, and cloud computing. Proficient in Python, SQL, R, C#, VBA, and JavaScript, with expertise in AWS, Azure, and Snowflake. Skilled in predictive analytics, and workflow automation using Databricks, Apache Airflow, and SSIS. Strong experience in database management with SQL Server and PostgreSQL and developing .NET applications. Proficient in data visualization using Tableau, Power BI, and QuickSight.

SKILLS

- **Programming & Scripting:** Python, SQL, C#, R, Java, JavaScript, VBA
- **BI & Analytics Tools:** Power BI, Tableau, QuickSight, Excel (Pivot Tables, VLOOKUP)
- **Database and Cloud Technologies:** SQL Server, Snowflake, PostgreSQL, DynamoDB, TibcoEBX, AWS (Lambda, Sagemaker, S3, EC2, Code pipeline, Kinesis, Aurora, Redshift), Azure (OpenAI, Desktop, Data Factory)
- **Big Data & Machine Learning:** Spark, Databricks, Apache Airflow, Hadoop, TensorFlow, PyTorch, XGBoost, LightGBM
- **Frameworks & Libraries:** ASP.NET, Dash, Chart.js, PySimpleGUI, Tkinter, Pandas, NumPy, SQLAlchemy
- **Miscellaneous:** OpenAI (GPT-3, GPT-4), Hugging Face, LLaMA3.2, Stable Diffusion, BERT, DALL·E, GitHub, GitLab

EXPERIENCE:

Data Analyst | University of Houston | Houston, TX **Jan 2023 - Present**

- Designed Tableau dashboards for teaching evaluation analytics, leveraging forecasting, What-If analysis, and parameterized views to visualize student participation, response distribution, and instructor performance trends.
- Processed and integrated large-scale survey data from MS SQL Server, applying LOD expressions, table calculations, and trend analysis to generate KPI monitoring, response rate insights, and department-level benchmarking.
- Managed and optimized ASP.NET web pages for course evaluations for the entire UH system, ensuring seamless updates and functionality while integrating interactive plots and reports using Chart.js for enhanced data visualization and user experience.
- Optimized over 50 T-SQL procedures, resulting in a 30% reduction in runtime and improved reporting performance.
- Continuing development of a GPT-powered chatbot using Azure OpenAI, trained on institutional datasets to assist professors and deans with integrating instructional tasks, improving user efficiency, and reducing 50% of manual work.
- Automated PDF report generation for 70 departments, leveraging Python, VBA, and Excel to save over 3,000 hours annually.

Data Science Analyst | Accenture (Client: HDFC Bank) | Hyderabad, India **Jun 2021 – Dec 2022**

- Designed interactive Power BI dashboards for customer segmentation and financial insights, leveraging DAX functions and Power Query for data transformation and real-time analytics.
- Built SQL-based KPI reports to analyze loan application conversion rates, payment default probabilities, and customer segmentation trends, optimizing financial risk models for HDFC Bank.
- Extracted, transformed and processed 10M+ HDFC customer records into PostgreSQL, applying validation rules & feature engineering for 1000+ features. Automated data ingestion via Python & SQL, enabling real-time insights in Power BI.
- Designed AWS Lambda ETL pipelines, processing model outputs for risk and fraud analytics, reducing processing time by 40%.
- Built machine learning models i.e. LightGBM to analyze home loan application risk, utilizing Python, H2O, and SQL to predict high-risk accounts and improve loan approval accuracy.
- Migrated terabytes of structured and semi-structured data from different storage layers like S3, and Snowflake to TibcoEbx platform, using Databricks and AWS Glue.

Junior Data Analyst | Mainavi Global (Client: Convergence Energy) | Hyderabad, India **Jun 2020 – Apr 2021**

- Developed real-time monitoring dashboards in Power BI, integrating live data ingestion using AWS Kinesis and Aurora to track solar energy production and consumption patterns.
- Assisted in designing and implementing a predictive model using Python, Pandas, and Scikit-learn to forecast solar energy generation and consumption, improving energy trading efficiency.
- Processed over 5 million rows of energy production data in PostgreSQL, identifying trends and anomalies.
- Designed automated ETL pipelines in Apache Airflow and AWS Glue, reducing data processing time by 40% and enhancing data reliability for energy analytics.

EDUCATION:

University of Houston, Cullen College of Engineering **Houston, TX**
Master of Science, Engineering Data Science; CGPA: 3.9/4 **Jan 2023 - Dec 2024**

Coursework: Machine Learning, Statistics, Big Data Analytics, Discrete Mathematics, Deep Learning, Text Mining, DBMT

National Institute of Technology Calicut **Kozhikode, India**
Bachelor of Technology, Electrical and Electronics Engineering; CGPA: 3.66/4 **Aug 2017 - May 2021**

PROJECTS:

Tableau Countries of The World Team Lead | Houston, TX [\[Code\]](#)

- Built Tableau dashboards analysing GDP, literacy, health, migration, and agriculture trends across countries.
- Identified key correlations in global development metrics through interactive visualizations.

YouTube Data Analytics Platform Team Lead | Houston, TX [\[Code\]](#)

- Processed and visualized YouTube analytics data using AWS S3, Glue, Lambda, and QuickSight to generate actionable insights.
- Automated data transformation workflows by converting CSV and JSON data to Parquet format using AWS Glue Spark jobs

CERTIFICATION AND ACCOMPLISHMENT:

- AWS Certified Solutions Architect – Associate, Microsoft Certified: Azure Fundamentals, and Tableau Developer.
- Secured Dean’s Merit Scholarship at the University of Houston for the 2023–2024 term.
- Team Lead for 6 research assistants at the Measurement and evaluation center (May 2024 – Present)
- Revamped course evaluation system for the entire UH system by adding additional questions to the existing system.